

★ WEEK 7 · PART 1 · PRACTICE

Cointegration, Worked by Hand

Find the Hedge Ratio, Then Tell It From Mere Correlation

The equilibrium spread and the cointegration test

Part 1's core idea, with the arithmetic shown. We estimate the hedge ratio that links two wandering prices into a steady spread, then see why two highly correlated series can still fail to cointegrate - the distinction that keeps a pairs trade from blowing up.

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PART 1 · APPLICATIONS

One Steady Spread From Two Drifters

Case A: estimate the cointegrating vector - the hedge ratio - and form the equilibrium spread, showing it stays flat while both prices climb. Case B: cointegration versus correlation, and why only one of two correlated pairs is tradable. Then a Claude Code lab.

What Part 1 Told Us

Now we apply it - with numbers

1

Each series wanders, the combination doesn't

Two prices can each be non-stationary, yet a particular linear combination of them is stationary - that's cointegration.

2

The cointegrating vector is a hedge ratio

The coefficient beta that makes the spread stationary is exactly how much of one asset to hold against the other.

3

Cointegration is not correlation

Two series can move together in the short run yet drift apart forever. Only a stationary spread supports a pairs trade.

Everything here is a calculator exercise: one regression, one spread, two comparisons.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from two price series.
- Estimate the spread, or compare two pairs.
- Then read whether it reverts.

What you should watch

- How beta turns two prices into one spread.
- How the spread stays bounded while prices climb.
- How correlation can mislead you.

The Hedge Ratio and the Spread

Two wandering prices, one steady link

Two related stocks each drift like a random walk, yet they stay tethered. Regressing one price on the other gives beta - the cointegrating vector - and the residual is the spread. We estimate beta by hand, form the spread, and confirm it hovers in a narrow band even as both prices climb ten points

The data

- Stock X drifts from 100 up to 112.
- Stock Y roughly tracks about twice X.
- Eight observations.

What we'll do by hand

- Regress Y on X for beta.
- Form the spread Y minus beta-X.
- Check that it stays bounded.

The Cointegrating Relationship

A stationary spread links two drifters

$$\underbrace{Y_t}_{I(1)} = \alpha + \beta \underbrace{X_t}_{I(1)} + \underbrace{s_t}_{\text{stationary spread}} \quad (\text{two wandering prices, one steady combination})$$

The pieces

- Both prices are individually non-stationary.
- Beta scales one to match the other.
- What's left over is the spread.

The magic

- The spread is stationary - it mean-reverts.
- So the two prices never wander far apart.
- Beta is the tether.

Estimate the Hedge Ratio

Regress one price on the other

$$\beta = \frac{\text{Cov}(Y,X)}{\text{Var}(X)} = \frac{33.23}{16.98} = \mathbf{1.96}, \quad \alpha = \bar{Y} - \beta\bar{X} = \mathbf{5.44}$$

The calculation

- Covariance of Y and X is 33.23.
- Variance of X is 16.98.
- Their ratio is beta = 1.96.

The meaning

- Hold one share of Y against 1.96 of X.
- That's the hedge ratio.
- The intercept sets the spread's level.

Form the Spread

Bounded while the prices climb

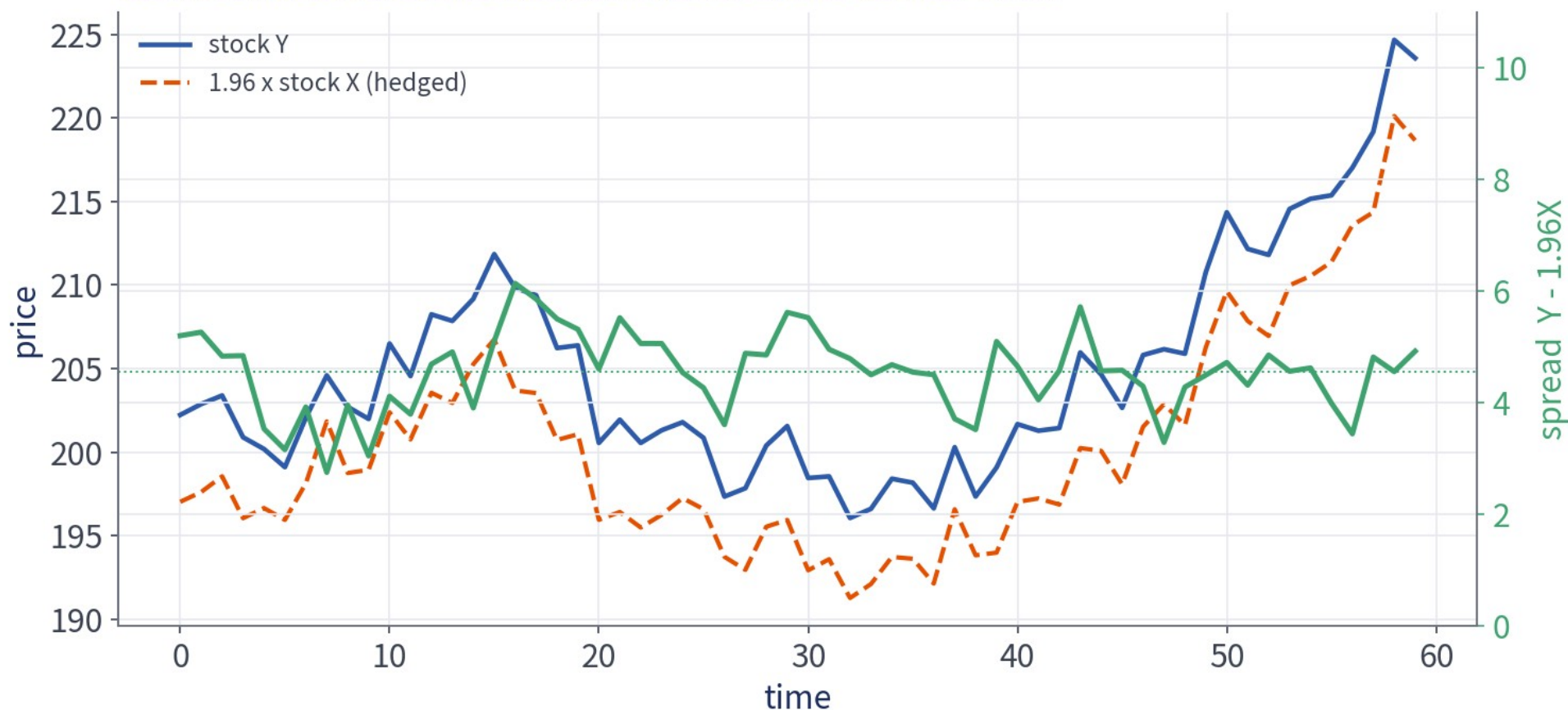
X	100	103	107	110	112
Y	201	208	215	221	224
spread $Y - 1.96X$	5.0	6.1	5.3	5.4	4.5

As X climbs from 100 to 112, the spread never leaves the 4.1-to-6.1 band, hovering near 5.2. Both prices trend, but their hedged combination is flat - the signature of cointegration.

Prices Climb, Spread Stays Flat

Stock Y and its hedged partner 1.96-times-X climb together over time, wandering up the chart. But the spread between them (green, right axis) barely moves - it mean-reverts inside a tight band. That flat green line is the equilibrium the two prices are tethered to

Case A: both prices wander and climb, but the spread stays flat and mean-reverts



Beta Is the Trade

From regression to hedge

What beta gives you

- The exact hedge ratio for a pairs trade.
- The spread you actually trade around.
- An equilibrium to bet on reverting to.

Why it matters

- When the spread stretches, you fade it.
- When it reverts, you profit.
- The whole trade rests on that stationarity.

Cointegration vs Correlation

Moving together is not enough

Two pairs of stocks both show a correlation near 0.997 - textbook lockstep. Yet only one is cointegrated. In the first, the spread reverts inside a bounded range; in the second, it drifts steadily wider and never comes back. Correlation looks the same for both; only the spread's behavior reveals which pair is tradable

The two pairs

- Both partners trend up strongly.
- Both correlate about 0.997.
- Indistinguishable on correlation alone.

What we'll do by hand

- Track each pair's spread over time.
- See one revert, the other drift.
- Read which is tradable.

The Crucial Distinction

Correlation is short-run, cointegration is long-run

correlation ≈ 0.997 for BOTH pairs, but only one has a stationary spread (cointegration $\neq$ correlation)

Correlation

- Measures short-run co-movement.
- Can be high even without a tether.
- Says nothing about drifting apart.

Cointegration

- Requires a stationary spread.
- Guarantees the gap stays bounded.
- That's what a pairs trade needs.

The Cointegrated Pair

A spread that reverts

time	1	2	3	4	5	6	7	
spread	+0.5	-0.8	+0.3	-0.6	+0.7	-0.4	+0.5	-0

The spread crosses zero again and again, staying inside a narrow band from -0.8 to +0.7. It reverts - a stationary spread - so the pair is tradable: fade the extremes, collect on the return.

What you see

- The spread flips sign every step.
- It never leaves the -0.8 to +0.7 band.
- A clear, bounded mean-reversion.

Why it is tradable

- Fade the spread when it stretches wide.
- Collect as it snaps back to zero.
- Stationary spread, repeatable edge.

The Correlated-Only Pair

A spread that drifts

time	1	2	3	4	5	6	7	
spread	0.0	0.6	1.3	2.1	3.0	4.2	5.5	6.9

Here the spread only grows - 0 to 6.9 - never turning back. Despite a 0.997 correlation, the gap wanders without bound: not cointegrated. A pairs trade here would keep losing as the spread widens.

What you see

- The spread climbs every single step.
- It runs from 0 all the way to 6.9.
- No sign of turning back.

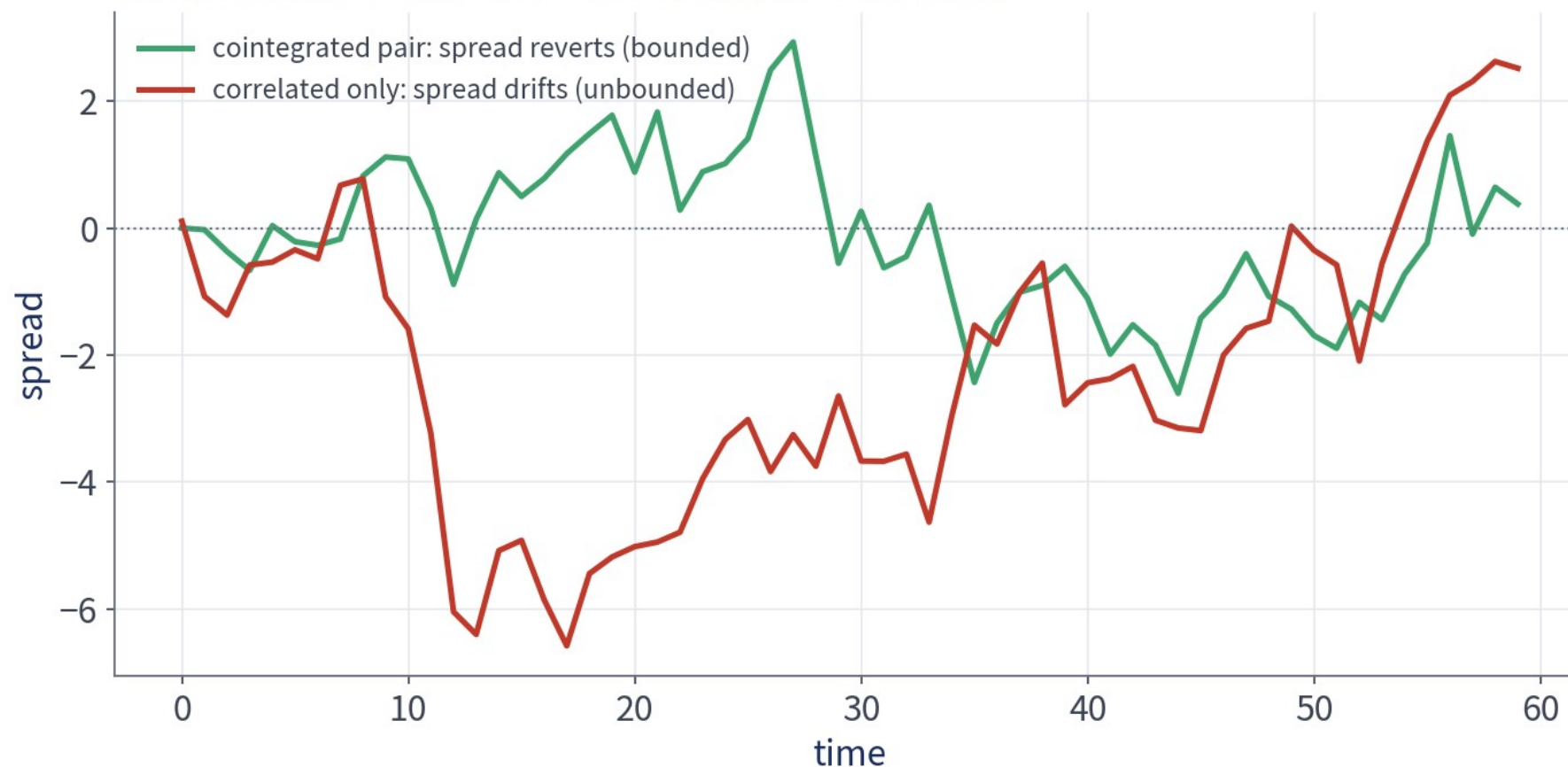
Why it is not tradable

- A short spread trade keeps losing.
- The gap widens without a mean to revert to.
- High correlation gave no warning.

One Reverts, One Runs Away

Both pairs correlate near 0.997, yet their spreads could not differ more. The green spread oscillates around zero - stationary, tradable. The red spread drifts steadily away and never returns - a random walk in disguise. Correlation could not tell them apart; the spread does.

Case B: both pairs correlate ~ 0.997 - but only the green spread reverts



Why the Distinction Saves You

Correlation is a trap here

The trap

- Pairing two stocks just because they correlate.
- Assuming the spread will always come back.
- Watching it drift and holding on.

The discipline

- Trade only cointegrated pairs.
- Test the spread for stationarity first.
- Correlation is necessary, not sufficient.

One Shared Trend Underneath

The anchor that makes it work

Cointegration means two prices share a single underlying random walk - a common trend. Beta scales one series so that shared trend cancels, leaving only a stationary spread. That's why the hedged combination stays flat, and why a genuine pairs trade rests on cointegration, not correlation: the shared trend is the anchor, and

$$Y_t = \gamma T_t + u_t, \quad X_t = T_t + v_t \Rightarrow Y_t - \gamma X_t = u_t - \gamma v_t \text{ (shared trend } T_t \text{ cancels)}$$

Reproduce the Spread and the Test

Direct Claude Code to check your arithmetic

GOAL Estimate the hedge ratio and spread, then distinguish cointegration from correlation.

[INPUT] Prompt · Claude Code

Write and run `part1_practice.py` that:

CASE A - hedge ratio:

- * regress $Y=[201,204,211,208,215,221,217,224]$ on $X=[100,102,105,103,107,110,108,112]$;
- * print beta, alpha, and the spread $Y - \text{beta} * X$.

CASE B - cointegration vs correlation:

- * build a reverting spread and a drifting spread;
- * print the correlation of each pair (both ~ 0.997);
- * run an ADF test on each spread.

Then plot the prices with the spread, and the two spreads side by side.

Check against the slides

- beta 1.96, alpha 5.44.
- spread in $[4.1, 6.1]$.
- both correlations ~ 0.997 .
- only the reverting spread is $I(0)$.

Deliverable: `part1_practice.py` whose printed numbers match the spread and the two pairs.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does the cointegrating vector beta represent?

**The hedge ratio
between the two
assets**

Q2 How do you know a pair is cointegrated?

**Their spread is
stationary - it mean-
reverts**

Q3 Why isn't high correlation enough?

**Correlated series can
still drift apart
forever**

All three come from the hedge-ratio regression and the cointegration-versus-correlation test.

Three Pitfalls to Avoid

Where cointegration trips people

Common mistakes

- "High correlation means a tradable pair."
- "Any linear combination gives the spread."
- "The spread will always come back."

The reality

- Correlation can hide a spread that drifts forever.
- Only the cointegrating beta makes it stationary.
- Test for stationarity - don't assume reversion.

Two Calculations to Remember

Part 1 applications in one screen

1

The hedge ratio makes the spread

Regressing Y on X gave $\beta = 1.96$ - the cointegrating vector. The spread $Y - 1.96X$ stayed in a tight 4.1-to-6.1 band while X climbed from 100 to 112. Beta is the hedge ratio, and the flat spread is the equilibrium you trade around.

2

Correlation is not cointegration

Two pairs both correlated near 0.997 behaved oppositely: one spread reverted inside a bounded range, the other drifted from 0 to 6.9 and never returned. Only a stationary spread supports a pairs trade - correlation alone would have misled you.

3

A shared trend is the anchor

Cointegration means the two prices ride one common random walk, which beta cancels to leave a stationary spread. That shared trend is what keeps the pair tethered - and what a genuine trade depends on.

Next — Part 2 practice: formally testing for cointegration with Engle-Granger and Johansen.